US012385403B2

(12) **United States Patent**
Hatta et al.

(10) **Patent No.: US 12,385,403 B2**
(45) **Date of Patent: Aug. 12, 2025**

(54) **TURBINE STATOR VANE**

(71) Applicant: **MITSUBISHI HEAVY INDUSTRIES, LTD.**, Tokyo (JP)

(72) Inventors: **Masayoshi Hatta**, Tokyo (JP); **Kazuki Morimoto**, Tokyo (JP); **Susumu Wakazono**, Tokyo (JP)

(73) Assignee: **MITSUBISHI HEAVY INDUSTRIES, LTD.**, Tokyo (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/702,088**

(22) PCT Filed: **Nov. 18, 2022**

(86) PCT No.: **PCT/JP2022/042830**
§ 371 (c)(1),
(2) Date: **Apr. 17, 2024**

(87) PCT Pub. No.: **WO2023/095721**
PCT Pub. Date: **Jun. 1, 2023**

(65) **Prior Publication Data**

US 2025/0012193 A1 Jan. 9, 2025

(30) **Foreign Application Priority Data**

Nov. 29, 2021 (JP) 2021-193277

(51) **Int. Cl.**
F01D 5/18 (2006.01)
F01D 9/02 (2006.01)

(52) **U.S. Cl.**
CPC ***F01D 5/18*** (2013.01); ***F01D 9/02*** (2013.01); *F05D 2260/201* (2013.01); *F05D 2300/611* (2013.01)

(58) **Field of Classification Search**
CPC F01D 5/18; F01D 5/187; F01D 5/288; F01D 9/02; F01D 9/04; F01D 9/041; F05D 2240/81; F05D 2300/611
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,628,880 A * 12/1971 Smuland F01D 9/04 415/115
4,012,167 A 3/1977 Noble
(Continued)

FOREIGN PATENT DOCUMENTS

JP 2001-254604 9/2001
JP 2004-60638 2/2004
(Continued)

OTHER PUBLICATIONS

International Search Report issued Dec. 20, 2022 in International Application No. PCT/JP2022/042830, with English-language Translation.

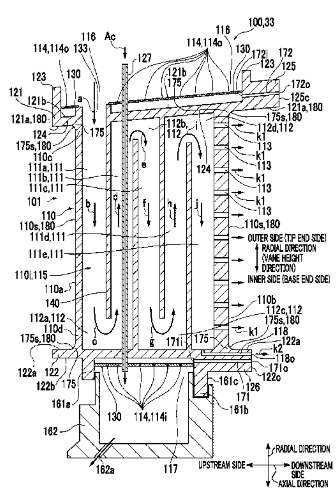
(Continued)

Primary Examiner — Courtney D Heinle
Assistant Examiner — Cameron A Corday
(74) *Attorney, Agent, or Firm* — Wenderoth, Lind & Ponack, L.L.P.

(57) **ABSTRACT**

A turbine stator vane according to at least one embodiment of the present disclosure comprises an airfoil, an inner shroud provided on the inner-peripheral side of the airfoil, and an outer shroud provided on the outer-peripheral side of the airfoil. The inner shroud has a first recess formed in the surface on the side opposite the airfoil across a gas path surface of the inner shroud. The outer shroud has a second recess formed in the surface on the side opposite the airfoil across a gas path surface of the outer shroud, and at least one outer passage that communicates with the second recess and does not communicate with the space inside the airfoil. The

(Continued)



number of outer passages is greater than the number of inner passages that communicate with the first recess in the inner shroud and do not communicate with the space inside the airfoil.

13 Claims, 5 Drawing Sheets

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,095,755	A *	8/2000	Houston	C23C 4/01 416/224
6,126,400	A *	10/2000	Nichols	F01D 5/288 415/177
10,544,685	B2	1/2020	Takamura et al.	
2004/0018082	A1*	1/2004	Soechting	F01D 9/041 415/115
2007/0148003	A1*	6/2007	Trishkin	F01D 5/18 416/241 R
2009/0074570	A1*	3/2009	Milleville	F01D 5/288 427/256
2015/0030461	A1*	1/2015	Mugglestone	F01D 9/065 415/115
2015/0285086	A1	10/2015	Bangerter	
2015/0322804	A1*	11/2015	Butler	F01D 9/041 29/889.71
2015/0322818	A1*	11/2015	Hannam	C23C 14/22 415/177
2015/0345307	A1*	12/2015	Davidson	F01D 5/3007 416/189
2018/0202301	A1	7/2018	Otomo et al.	
2019/0032499	A1*	1/2019	Matsuo	F01D 25/12
2022/0186623	A1	6/2022	Koyabu et al.	

FOREIGN PATENT DOCUMENTS

JP	2015-507128	3/2015
JP	2017-36710	2/2017
JP	2017-160892	9/2017
JP	6344869	6/2018
WO	2020/213381	10/2020

OTHER PUBLICATIONS

Translation of the International Preliminary Report on Patentability issued Jun. 13, 2024 in International Application No. PCT/JP2022/042830.

* cited by examiner

FIG. 1

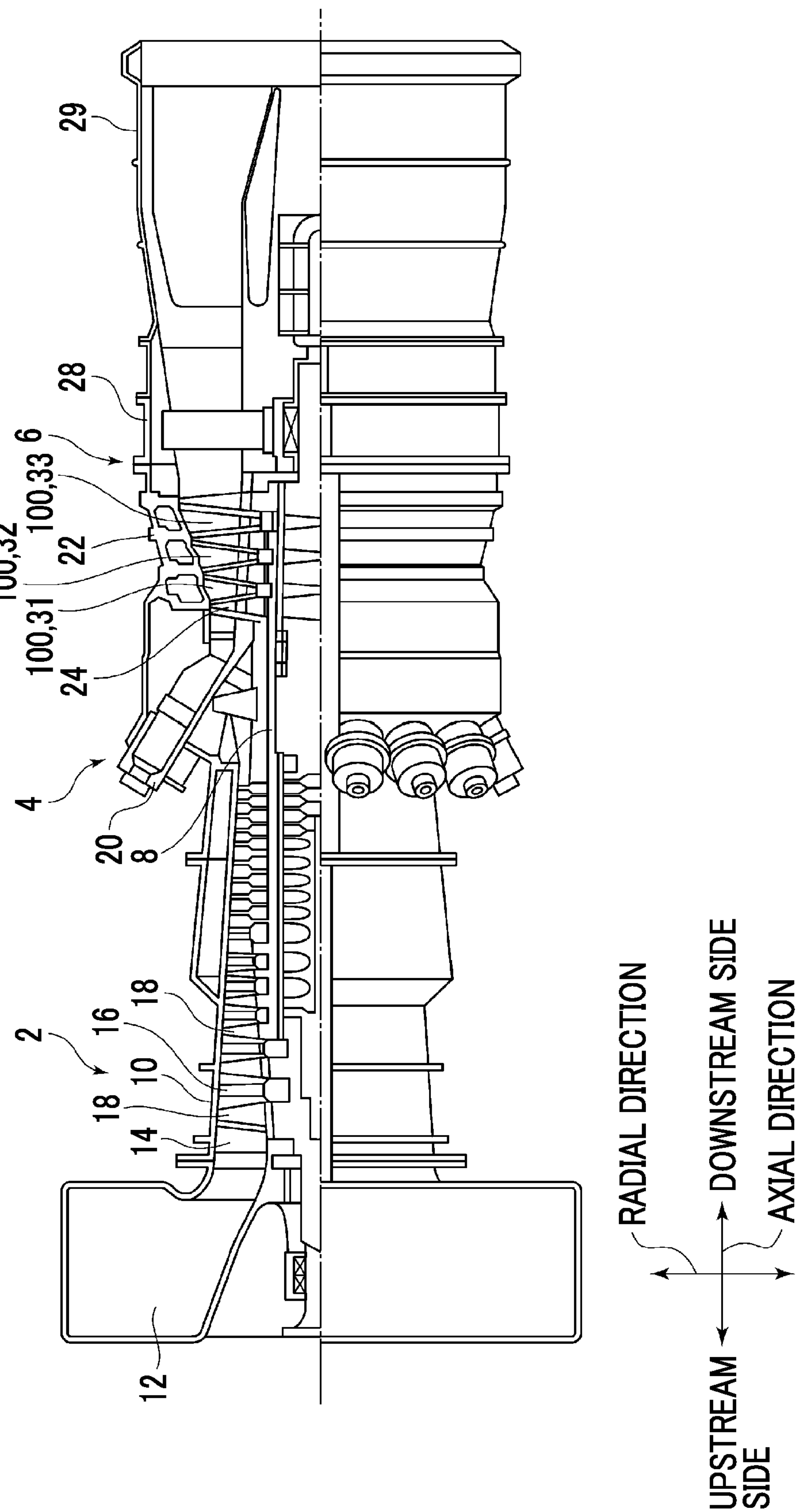

1
29
28
6
100,33
22
100,32
100,31
24
4
20
8
2
18
16
10
18
14
12
RADIAL DIRECTION
DOWNSTREAM SIDE
AXIAL DIRECTION
UPSTREAM SIDE

FIG. 2

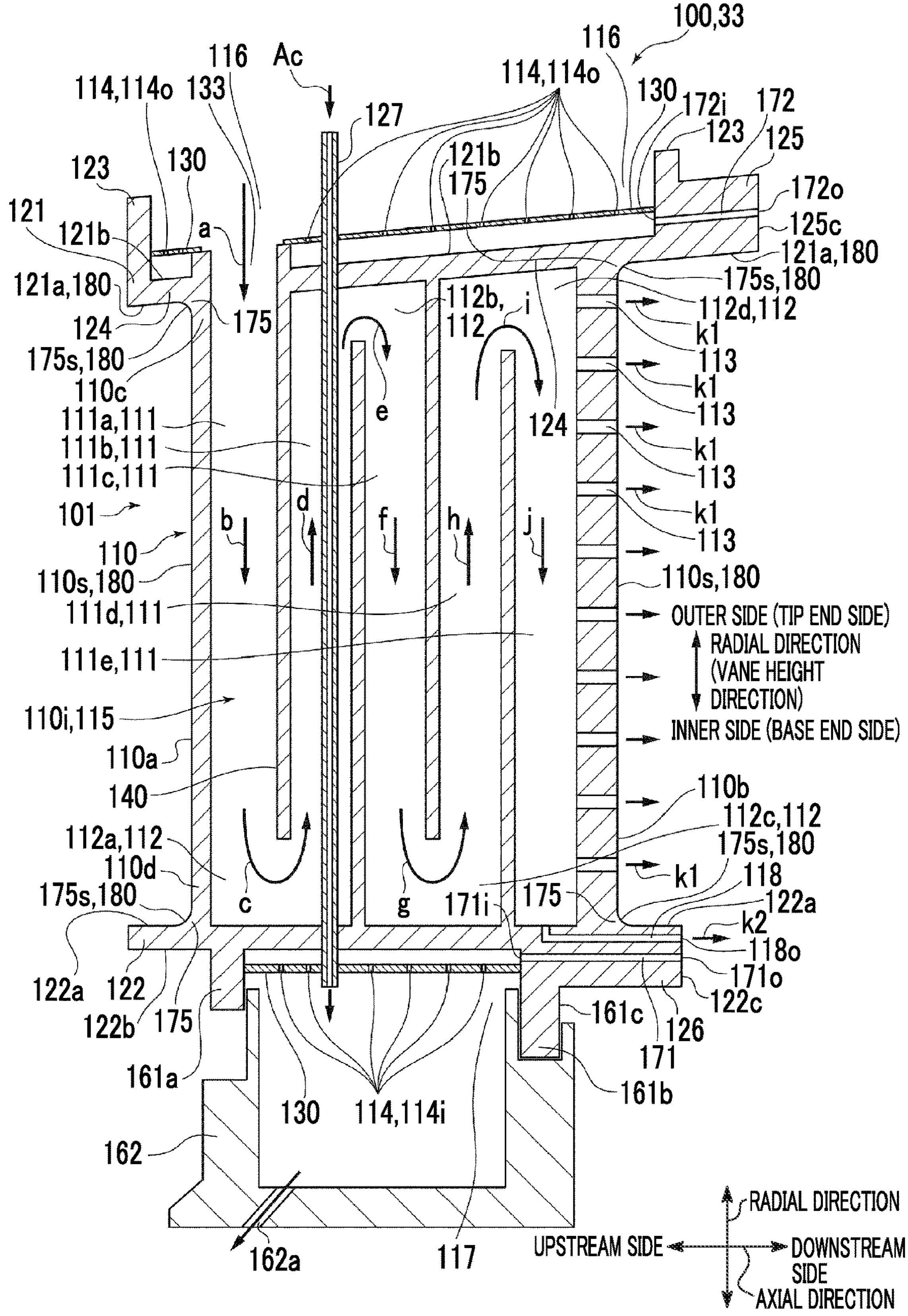

100,33
116
114,114o
130
172i
172
123
125
172o
125c
121a,180
175s,180
112d,112
k1
113
124
116
Ac
127
114,114o
133
123
130
121
121b
121a,180
124
175s,180
110c
a
175
121b
175
112b,
112
i
e
111a,111
111b,111
111c,111
101
110
110s,180
b
d
f
h
j
110s,180
111d,111
111e,111
OUTER SIDE (TIP END SIDE)
RADIAL DIRECTION
(VANE HEIGHT
DIRECTION)
INNER SIDE (BASE END SIDE)
110i,115
110a
140
112a,112
110d
175s,180
c
g
171i
175
110b
112c,112
175s,180
118
122a
k2
118o
171o
122c
122a
122
122b
175
161a
130
114,114i
161c
126
171
161b
162
162a
117
RADIAL DIRECTION
UPSTREAM SIDE
DOWNSTREAM
SIDE
AXIAL DIRECTION

FIG. 3

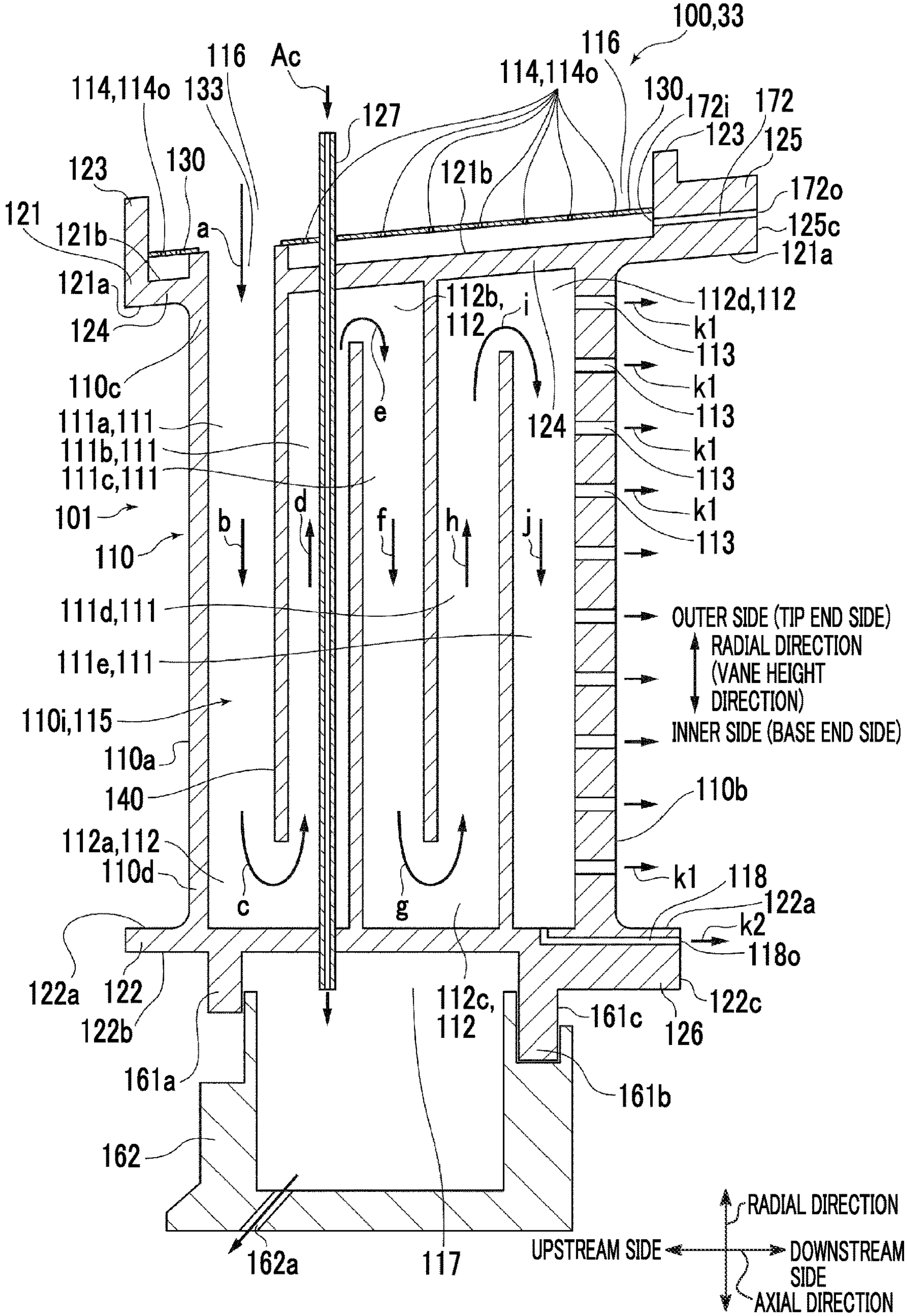

100,33
116
114,114o
130
172i
172
123
125
172o
125c
121a
116
Ac
127
114,114o
133
123
130
121b
121
121b
121a
124
a
112b,
112
i
112d,112
k1
113
110c
e
124
111a,111
111b,111
111c,111
101
110
b
d
f
h
j
111d,111
111e,111
OUTER SIDE (TIP END SIDE)
RADIAL DIRECTION
(VANE HEIGHT
DIRECTION)
INNER SIDE (BASE END SIDE)
110i,115
110a
140
110b
112a,112
110d
c
g
118
122a
k2
118o
122a
122
122b
112c,
112
122c
126
161c
161a
161b
162
162a
117
RADIAL DIRECTION
UPSTREAM SIDE
DOWNSTREAM
SIDE
AXIAL DIRECTION

FIG. 4

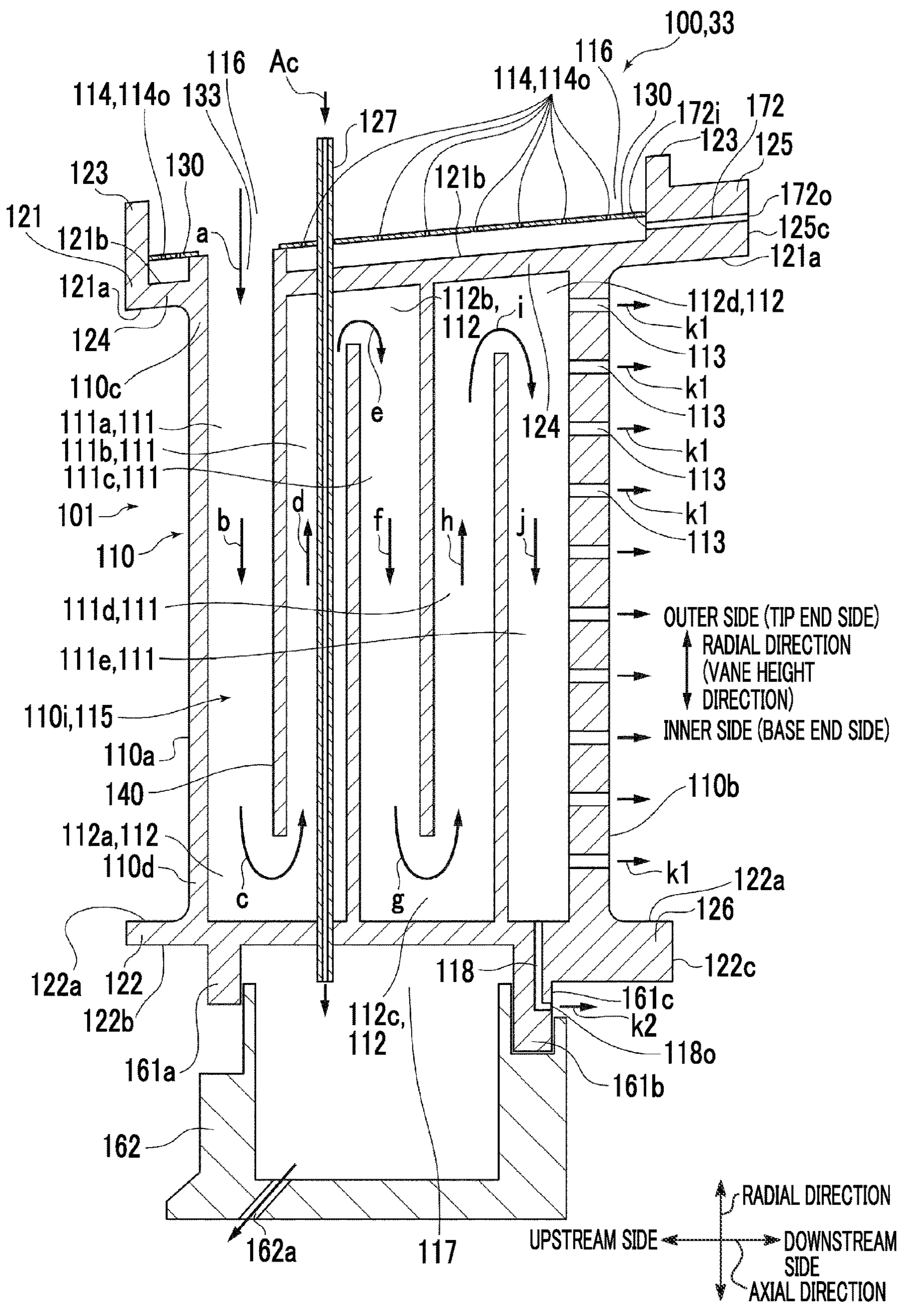

100,33
116
114,114o
130
172i
172
123
125
172o
125c
121a
121b
127
Ac
116
133
114,114o
130
123
121
121b
121a
124
110c
a
112b,
112
i
112d,112
k1
113
e
124
111a,111
111b,111
111c,111
101
110
b
d
f
h
j
111d,111
111e,111
OUTER SIDE (TIP END SIDE)
RADIAL DIRECTION
(VANE HEIGHT
DIRECTION)
INNER SIDE (BASE END SIDE)
110i,115
110a
140
110b
112a,112
110d
c
g
122a
126
122c
122a
122
122b
118
161c
k2
118o
112c,
112
161a
161b
162
162a
117
RADIAL DIRECTION
UPSTREAM SIDE
DOWNSTREAM
SIDE
AXIAL DIRECTION

FIG. 5

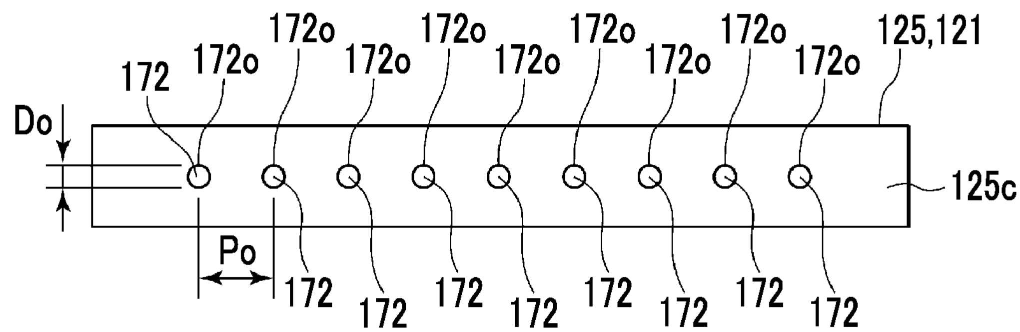
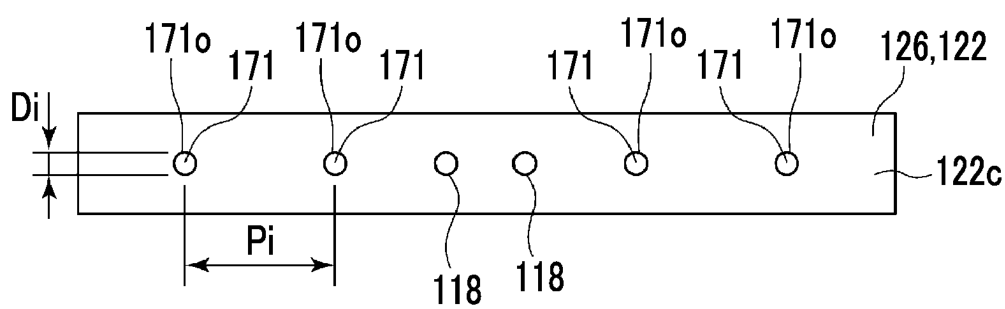

172
172o
172o
172o
172o
172o
172o
172o
172o
172o
125,121
Do
125c
Po
172
172
172
172
172
172
172
172

FIG. 6


171o
171
171o
171
171
171o
171
171o
126,122
Di
122c
Pi
118
118

TURBINE STATOR VANE

TECHNICAL FIELD

The present disclosure relates to a turbine stator vane.

The present application claims priority based on Japanese Patent Application No. 2021-193277 filed in Japan on Nov. 29, 2021, the contents of which are incorporated herein by reference.

BACKGROUND ART

For example, a turbine stator vane used in a gas turbine or the like is exposed to a high-temperature fluid such as combustion gas, and thus has a structure for cooling. For example, in a turbine stator vane described in PTL 1, a stator vane body (airfoil portion), an inner shroud, and an outer shroud are respectively cooled with cooling air (refer to PTL 1).

CITATION LIST

Patent Literature

[PTL 1] International Publication No. WO2020/213381

SUMMARY OF INVENTION

Technical Problem

When the temperature of the combustion gas in the actual machine was measured, it was found that the temperature of the combustion gas on the hub side was lower than a predetermined assumption, and it was found that the metal temperature of the inner shroud was also lower than the predetermined assumption. In a case where the metal temperature of the turbine stator vane is lower than the temperature assumed in advance, the efficiency of the gas turbine is reduced. Therefore, it is desirable that the metal temperature of the turbine stator vane is a temperature close to the temperature assumed in advance.

At least one embodiment of the present disclosure is made in view of the above circumstances, and an object thereof is to optimize the cooling of a turbine stator vane to improve the turbine efficiency.

Solution to Problem

(1) According to at least one embodiment of the present disclosure, there is provided a turbine stator vane including:

an airfoil portion;

an inner shroud that is provided on an inner peripheral side of the airfoil portion; and

an outer shroud that is provided on an outer peripheral side of the airfoil portion,

wherein the inner shroud includes a first recessed portion that is formed on a surface on a side opposite to the airfoil portion by interposing a gas path surface of the inner shroud,

the outer shroud includes a second recessed portion that is formed on a surface on a side opposite to the airfoil portion by interposing a gas path surface of the outer shroud, and at least one outer passage communicating with the second recessed portion and not communicating with a space inside the airfoil portion, and

the number of the outer passages is greater than the number of inner passages communicating with the first recessed portion in the inner shroud and not communicating with the space inside the airfoil portion.

Advantageous Effects of Invention

According to at least one embodiment of the present disclosure, it is possible to optimize the cooling of the turbine stator vane and improve the turbine efficiency.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a schematic configuration diagram showing a gas turbine **1** according to one embodiment in which turbine stator vanes according to some embodiments are used.

FIG. **2** is a schematic internal sectional view of the turbine stator vane according to the embodiment.

FIG. **3** is a schematic internal sectional view of a turbine stator vane according to another embodiment.

FIG. **4** is a schematic internal sectional view of a turbine stator vane according to still another embodiment.

FIG. **5** is a view of an end surface of an outer shroud on a trailing edge side in the turbine stator vane shown in FIGS. **2** to **4** as seen from a downstream side.

FIG. **6** is a view of an end surface of an inner shroud on a trailing edge side in the turbine stator vane shown in FIG. **2** as seen from a downstream side.

DESCRIPTION OF EMBODIMENTS

Hereinafter, some embodiments of the present disclosure will be described with reference to the accompanying drawings. Dimensions, materials, shapes, relative arrangements, and the like of components described as embodiments or illustrated in the drawings are not intended to limit the scope of the present disclosure, but are merely explanatory examples.

For example, an expression representing a relative or absolute arrangement such as "in a certain direction", "along a certain direction", "parallel", "orthogonal", "center", "concentric", or "coaxial" does not strictly represent only such an arrangement, but also a tolerance or a state of being relatively displaced with an angle or a distance to the extent that the same function can be obtained.

For example, expressions such as "identical", "equal", and "homogeneous" indicating that things are in an equal state does not strictly represent only the equal state, but also a tolerance or a state where there is a difference to the extent that the same function can be obtained.

For example, an expression representing a shape such as a quadrangular shape or a cylindrical shape does not represent only a shape such as a quadrangular shape or a cylindrical shape in a geometrically strict sense, but also a shape including an uneven portion, a chamfered portion, and the like within a range in which the same effect can be obtained.

Meanwhile, the expressions "being provided with", "being prepared with", "comprising", "including", or "having" one component are not exclusive expressions excluding the presence of other components.

FIG. **1** is a schematic configuration diagram showing a gas turbine **1** according to one embodiment in which turbine stator vanes according to some embodiments are used.

As shown in FIG. **1**, the gas turbine **1** according to one embodiment includes a compressor **2** for generating compressed air, a combustor **4** for generating combustion gas using compressed air and fuel, and a turbine **6** configured to be rotationally driven by the combustion gas. In a case of the gas turbine **1** for power generation, a generator (not illustrated) is connected to the turbine **6**, and power generation is performed by rotational energy of the turbine **6**.

Specific configuration examples of each part in the gas turbine **1** will be described with reference to FIG. **1**.

The compressor **2** includes a compressor casing **10**, an air intake port **12** provided on an inlet side of the compressor casing **10** to take in air, a rotor shaft **8** provided to penetrate both the compressor casing **10** and a turbine casing **22** to be described later, and various vanes disposed in the compressor casing **10**. The various vanes include an inlet guide vane **14** provided on the air intake port **12** side, a plurality of compressor stator vanes **16** fixed to the compressor casing **10** side, and a plurality of compressor rotor vanes **18** embedded in the rotor shaft **8** to be alternately arranged in an axial direction with respect to the compressor stator vanes **16**. The compressor **2** may include other constituent elements such as a bleed air chamber (not illustrated). Such a compressor **2** causes the air taken in from the air intake port **12** to pass through the plurality of compressor stator vanes **16** and the plurality of compressor rotor vanes **18** and compresses the air to generate compressed air. Then, the compressed air is sent from the compressor **2** to the combustor **4** on the downstream side.

The combustor **4** is disposed in a casing (combustor casing) **20**. As shown in FIG. **1**, a plurality of combustors **4** may be disposed in an annular shape around the rotor shaft **8** in the casing **20**. The combustor **4** is supplied with the fuel and compressed air generated by the compressor **2**, and combusts the fuel to generate a high-temperature and high-pressure combustion gas G which is a working fluid of the turbine **6**. Then, the combustion gas is sent from the combustor **4** to the turbine **6** in a rear stage.

The turbine **6** includes the turbine casing **22** and the various turbine vanes disposed in the turbine casing **22**. The various turbine vanes include a plurality of turbine stator vanes **100** fixed to the turbine casing **22** side, and a plurality of turbine rotor vanes **24** planted in the rotor shaft **8** to be alternately arranged in the axial direction with respect to the turbine stator vanes **100**.

In the turbine **6**, the rotor shaft **8** extends in the axial direction (right-left direction in FIG. **1**), and the combustion gas flows from the combustor **4** side to the exhaust casing **28** side (from the left side to the right side in FIG. **1**). Therefore, in FIG. **1**, the left side shown in the drawing is an axial upstream side, and the right side illustrated in the drawing is an axial downstream side. In addition, in the following description, in a case where a radial direction is simply described, it represents the same direction as a radial direction orthogonal to the rotor shaft **8**.

In the gas turbine **1** according to the embodiment, the turbine stator vanes **100** include first-stage stator vanes **31**, second-stage stator vanes **32**, and third-stage stator vanes **33** provided in order from the axial upstream side.

The turbine rotor vane **24** is configured to generate a rotational driving force from the high-temperature and high-pressure combustion gas flowing in the turbine casing **22** together with the turbine stator vane **100**. The rotational driving force is transmitted to the rotor shaft **8**, so that the generator connected to the rotor shaft **8** is driven.

An exhaust chamber **29** is connected to the turbine casing **22** on the axial downstream side through the exhaust casing **28**. The combustion gas after driving of the turbine **6** is discharged to the outside through the exhaust casing **28** and the exhaust chamber **29**.

FIG. **2** is a schematic internal sectional view of the turbine stator vane **100** of the embodiment, and shows a cross section along a camber line of an airfoil portion.

FIG. **3** is a schematic internal sectional view of the turbine stator vane **100** of another embodiment, and shows a cross section along the camber line of the airfoil portion.

FIG. **4** is a schematic internal sectional view of the turbine stator vane **100** of still another embodiment, and shows a cross section along the camber line of the airfoil portion.

Hereinafter, the structure of the turbine stator vane **100** according to some embodiments will be described. In the following description, the structure of the third-stage stator vane **33** will be described. However, the first-stage stator vane **31** or the second-stage stator vane **32** may have the same structure as the third-stage stator vane **33**. In addition, in a case where the gas turbine **1** according to the embodiment has the turbine stator vane **100** on the axial downstream side of the third-stage stator vane **33**, the turbine stator vane **100** may have the same structure as the third-stage stator vane **33**.

As shown in FIGS. **2** to **4**, the turbine stator vane **100** according to some embodiments includes a vane body **101**, an impingement plate **130**, and an air pipe **127**.

The vane body **101** according to some embodiments includes an airfoil portion **110** having a plurality of cooling channels **111** separated by partition walls **140**, an outer shroud **121** provided on a tip end **110***c* side of the airfoil portion **110**, that is, on a radial outer side, and an inner shroud **122** provided on a base end **110***d* side (base end side) of the airfoil portion **110**, that is, on a radial inner side. In the following description, the radial direction is referred to as a vane height direction of the airfoil portion **110** or simply as a vane height direction. In addition, for convenience of description, the plurality of cooling channels **111** will be referred to as a first cooling channel **111***a*, a second cooling channel **111***b*, a third cooling channel **111***c*, a fourth cooling channel **111***d*, and a fifth cooling channel **111***e* in order from a leading edge **110***a* side to a trailing edge **110***b* side of the airfoil portion **110**. However, in the following description, in a case where there is no need to distinguish the cooling channels **111***a*, **111***b*, **111***c*, **111***d*, and **111***e* from each other, the description of the alphabet after the number in the reference numeral may be omitted, and the cooling channels may be simply referred to as a cooling channel **111**.

In the turbine stator vane **100** according to some embodiments, the first cooling channel **111***a* and the second cooling channel **111***b* are separated from each other by the partition wall **140**, and the second cooling channel **111***b* and the third cooling channel **111***c* are separated from each other by the partition wall **140**. The third cooling channel **111***c* and the fourth cooling channel **111***d* are separated from each other by the partition wall **140**, and the fourth cooling channel **111***d* and the fifth cooling channel **111***e* are separated from each other by the partition wall **140**.

In the turbine stator vane **100** according to some embodiments, a gas path surface is a surface with which the combustion gas comes into contact in a case where the turbine stator vane **100** according to some embodiments is disposed in the turbine, and corresponds to outer surfaces **121***a* and **122***a* of the outer shroud **121** and the inner shroud **122** shown in FIGS. **2** to **4**. In the turbine stator vane **100** according to some embodiments, the airfoil portion **110** and the shrouds **121** and **122** are manufactured by, for example, casting.

For example, in the turbine stator vane **100** according to some embodiments shown in FIGS. **2** to **4**, four turning channels **112** are formed. Specifically, in order from a leading edge **110**a side, a first turning channel **112**a communicates with the first cooling channel **111**a and the second cooling channel **111**b, and a second turning channel **112**b communicates with the second cooling channel **111**b and the third cooling channel **111**c. A third turning channel **112**c communicates with the third cooling channel **111**c and the fourth cooling channel **111**d, and a fourth turning channel **112**d communicates with the fourth cooling channel **111**d and the fifth cooling channel **111**e.

In the turbine stator vane **100** according to some embodiments shown in FIGS. **2** to **4**, a plurality of cooling holes **113** are formed in the vicinity of the trailing edge **110**b of the airfoil portion **110**, the cooling holes **113** communicate with the fifth cooling channel **111**e on the upstream side in a flow direction of cooling medium, and the downstream side of the cooling holes **113** is open to an end portion of the trailing edge **110**b.

In addition, in the turbine stator vane **100** according to some embodiments shown in FIGS. **2** and **3**, in the inner shroud **122**, at least one outlet passage **118** is formed, which communicates with the fifth cooling channel **111**e on the upstream side in the flow direction of the cooling medium and is open on the downstream side to, for example, an end surface **122**c of the inner shroud **122** on the trailing edge side.

In the turbine stator vane **100** according to the embodiment shown in FIG. **4**, in the inner shroud **122**, at least one outlet passage **118** is formed, which communicates with the fifth cooling channel **111**e on the upstream side in the flow direction of the cooling medium and is open on the downstream side to, for example, an end surface **161**c of a downstream rib **161**b on a trailing edge side, which will be described later.

In the turbine stator vane **100** according to some embodiments shown in FIGS. **2** to **4**, a serpentine channel **115** including the plurality of cooling channels **111** and the plurality of turning channels **112** is formed.

In the turbine stator vane **100** according to some embodiments shown in FIGS. **2** to **4**, the outer shroud **121** is formed with a bottom portion **124** forming the gas path surface and an outer wall portion **123** extending from both ends of the bottom portion **124** in the axial direction and the circumferential direction to a side opposite to the gas path surface in the vane height direction. The impingement plate **130** fixed to the outer wall portion **123** and having a plurality of through-holes **114** (outer impingement hole **114**o) is disposed on the outer shroud **121**.

The outer shroud **121** is formed with at least one outer passage **172** that communicates with the space between the impingement plate **130** and an inner surface **121**b of the outer shroud **121** and the outside (the outside of the airfoil portion **110**) of the turbine stator vane **100**, of the second recessed portion (that is, the internal space **116** of the outer shroud **121**) surrounded by the outer wall portion **123**.

In the turbine stator vane **100** according to some embodiments shown in FIGS. **2** to **4**, at least one outer passage **172** is formed in a trailing edge side end portion **125** on the axial downstream side with respect to the internal space **116**.

In the turbine stator vane **100** according to some embodiments shown in FIGS. **2** to **4**, an outlet end **172**o of the outer passage **172** is open to, for example, an end surface **125**c of the outer shroud **121** on the trailing edge side.

In the turbine stator vane **100** according to some embodiments shown in FIGS. **2** to **4**, the inner shroud **122** is formed with the upstream rib **161**a that protrudes inward from an inner surface **122**b of the inner shroud **122** in the vane height direction and is disposed on the leading edge **110**a side and the downstream rib **161**b that protrudes inward from the inner surface **122**b of the inner shroud **122** in the vane height direction and is disposed on the trailing edge **110**b side.

In the turbine stator vane **100** shown in FIG. **2**, the impingement plate **130** partitioning the internal space **117** and having the plurality of through-holes **114** (inner impingement hole **114**i) is disposed between the upstream rib **161**a and the downstream rib **161**b. In the turbine stator vane **100** shown in FIGS. **3** and **4**, the impingement plate **130** is not disposed.

In the turbine stator vane **100** shown in FIG. **2**, the inner shroud **122** is formed with at least one inner passage **171** that communicates with the space between the impingement plate **130** and the inner surface **122**b of the inner shroud **122** and the outside (the outside of the airfoil portion **110**) of the turbine stator vane **100**, of the first recessed portion (that is, the internal space **117** of the inner shroud **122**) between the upstream rib **161**a and the downstream rib **161**b.

In the turbine stator vane **100** shown in FIG. **2**, at least one inner passage **171** is formed in a trailing edge side end portion **126** on the axial downstream side with respect to the internal space **117**. In FIG. **2**, for convenience of illustration, the inner passage **171** and the outlet passage **118** are illustrated as being shifted in the vane height direction. However, as illustrated in FIG. **6** to be described later, the inner passage **171** and the outlet passage **118** may be formed at the same position in the vane height direction.

In the turbine stator vane **100** shown in FIG. **2**, the inner passage **171** does not communicate with a space **110**i inside the airfoil portion **110**. Here, the fact that the inner passage **171** does not communicate with the space **110**i inside the airfoil portion **110** means that the direct connection destination of the inner passage **171** is not the space **110**i inside the airfoil portion **110**. That is, the fact that the inner passage **171** does not communicate with the space **110**i inside the airfoil portion **110** means that all of the end portion openings (an inlet end **171**i and an outlet end **171**o) of the inner passage **171** do not face the space **110**i inside the airfoil portion **110**.

In some embodiments, the space **110**i inside the airfoil portion **110** is a void formed inside the airfoil portion **110** (a region between the outer surface **121**a of the outer shroud **121** and the outer surface **122**a of the inner shroud **122**), and includes, for example, the serpentine channel **115**.

In the turbine stator vane **100** shown in FIGS. **3** and **4**, the inner passage **171** is not formed.

The turbine stator vane **100** according to some embodiments shown in FIGS. **2** to **4** includes, for example, the air pipe **127** penetrating the airfoil portion **110** in the vane height direction in the second cooling channel **111**b. One end of the air pipe **127** is open to an internal space **117** formed in a holding ring **162** supported by the inner shroud **122**. The holding ring **162** is supported by the inner shroud **122** through the upstream rib **161**a and the downstream rib **161**b of the inner shroud **122**.

In addition, the holding ring **162** includes, for example, a circulation hole **162**a in the bottom surface.

For example, the cooling medium supplied to the turbine stator vane **100** is the compressed air drawn from the compressor **2**.

In the turbine stator vane **100** according to some embodiments shown in FIGS. **2** to **4**, the cooling medium supplied to the serpentine channel **115** is supplied from the outside to the internal space **116** of the outer shroud **121** as indicated by an arrow a. The cooling medium flows into the first cooling channel **111**a through an opening **133** formed in the outer shroud **121**, and as indicated by an arrow b, flows in the first cooling channel **111**a from the tip end **110**c side toward the base end **110**d side along the vane height direction. Thereafter, the cooling medium flowing into the first cooling channel **111**a sequentially flows through each turning channel **112** and each cooling channel **111** as indicated by arrows c to j. In this way, the cooling medium flows from the leading edge **110**a side toward the trailing edge **110**b side in the airfoil portion **110** in the same direction as the main flow of the combustion gas.

The cooling medium flowing into the fifth cooling channel **111**e is discharged from a plurality of cooling holes **113** open to the trailing edge **110**b to the outside of the airfoil portion **110** in the combustion gas as indicated by an arrow k**1**.

In addition, the cooling medium flowing into the fifth cooling channel **111**e is discharged into the combustion gas outside the airfoil portion **110** through the outlet passage **118** as indicated by an arrow k**2**.

In addition, in the case of the turbine stator vane **100** according to some embodiments shown in FIGS. **2** to **4**, the cooling medium supplied from the outside into a region (internal space **116**) on the radial outer side (tip end **110**c side) with respect to the impingement plate **130** is blown to the inner surface **121**b of the bottom portion **124** of the outer shroud **121** on the radial outer side (tip end **110**c side) through the plurality of through-holes **114** of the impingement plate **130** disposed on the outer shroud **121**. The cooling medium impingement cools (collision cools) the inner surface **121**b. Accordingly, the bottom portion **124** of the outer shroud **121** can be cooled by the cooling medium.

The cooling medium that cools the bottom portion **124** of the outer shroud **121** is discharged from the second recessed portion (internal space **116**) into the combustion gas outside the airfoil portion **110** through the plurality of outer passages **172**.

The cooling medium Ac supplied from the internal space **116** of the outer shroud **121** is supplied to the internal space **117** formed in the holding ring **162** on the inner shroud **122** side through the air pipe **127**.

In the turbine stator vane **100** shown in FIG. **2**, a portion of the cooling medium Ac is applied as cooling air for impingement cooling (collision cooling) to the inner surface **122**b of the inner shroud **122** through the plurality of through-holes **114** of the impingement plate **130** disposed on the inner shroud **122**.

In the turbine stator vane **100** shown in FIG. **2**, the remaining cooling medium Ac is supplied from the circulation hole **162**a to the inter-stage cavity (not illustrated), and the combustion gas is prevented from flowing back to an inter-stage cavity as a purge air.

In the turbine stator vane **100** shown in FIG. **2**, the cooling medium Ac that cools the inner surface **122**b of the inner shroud **122** is discharged from the first recessed portion (internal space **117**) to the combustion gas outside the airfoil portion **110** through the inner passage **171**.

In the turbine stator vane **100** shown in FIGS. **3** and **4**, the cooling medium Ac supplied to the internal space **117** is supplied from the circulation hole **162**a to the inter-stage cavity (not illustrated), and the combustion s is prevented from flowing back to the inter-stage cavity as the purge air.

As described above, the turbine stator vane **100** according to some embodiments shown in FIGS. **2** to **4** includes the airfoil portion **110**, the inner shroud **122** provided on an inner peripheral side of the airfoil portion **110**, and the outer shroud **121** provided on an outer peripheral side of the airfoil portion **110**.

In the turbine stator vane **100** according to some embodiments shown in FIGS. **2** to **4**, the inner shroud **122** has the first recessed portion (internal space **117**) formed on a surface on a side opposite to the airfoil portion **110** with the gas path surface (outer surface **122**a) of the inner shroud **122** interposed therebetween.

In the turbine stator vane **100** according to some embodiments shown in FIGS. **2** to **4**, the outer shroud **121** includes the second recessed portion (internal space **116**) formed on a surface on a side opposite to the airfoil portion **110** with the gas path surface (outer surface **121**a) interposed therebetween, and at least one outer passage **172** communicating with the second recessed portion (internal space **116**) and not communicating with the space **110**i inside the airfoil portion **110**.

Here, the fact that the outer passage **172** does not communicate with the space **110**i inside the airfoil portion **110** means that the direct connection destination of the outer passage **172** is not the space **110**i inside the airfoil portion **110**. That is, the fact that the outer passage **172** does not communicate with the space **110**i inside the airfoil portion **110** means that all of the end portion openings (an inlet end **172**i and an outlet end **172**o) of the outer passage **172** do not face the space **110**i inside the airfoil portion **110**.

When the temperature of the combustion gas in the actual machine of the gas turbine **1** was measured by the inventors, it was found that the temperature of the combustion gas on the hub side in the third-stage stator vane **33** was lower than a predetermined assumption, and it was found that the metal temperature of the inner shroud **122** in the third-stage stator vane **33** was also lower than a predetermined assumption. In a case where the metal temperature of the turbine stator vane **100** is lower than the temperature assumed in advance, the efficiency of the gas turbine **1** is reduced. Therefore, it is desirable that the metal temperature of the turbine stator vane **100** is a temperature close to the temperature assumed in advance.

Therefore, in the turbine stator vane **100** of some embodiments, the number of the outer passages **172** is set to be greater than the number of the inner passages **171**.

Therefore, cooling capacity of the inner shroud **122** can be suppressed to be lower than cooling capacity of the outer shroud **121**, and thus excessive cooling of the inner shroud **122** can be suppressed, and the cooling of the turbine stator vane **100** can be optimized. Accordingly, the turbine efficiency can be improved.

More specifically, in some embodiments, the turbine stator vane **100** may be configured to satisfy any one of the following (A), (B), or (C) conditions.

(A) The inner shroud **122** does not have the inner passage **171**.

(B) The inner shroud **122** has one inner passage **171**, and the cross-sectional area Si of the outlet end **171**o of the one inner passage **171** is smaller than the total ΣSo of the cross-sectional areas So of the outlet ends **172**o of the at least one outer passage **172**.

(C) The inner shroud **122** has two or more inner passages **171**, and a value (Pi/Di) obtained by dividing an interval Pi between the inner passages **171** by the diameter Di of the inner passage **171** is larger than a value (Po/Do) obtained by dividing an interval Po between the outer passages **172** by the diameter Do of the outer passage **172**.

The condition (A) will be described.

For example, as shown in FIGS. **3** and **4**, the condition (A) is satisfied in a case where the impingement plate **130** is not disposed in the internal space **117**, the inner passage **171** is not formed in the turbine stator vane **100**, and the inner surface **122***b* of the inner shroud **122** is not impingement cooled.

In the turbine stator vane **100** shown in FIGS. **3** and **4**, the inner surface **122***b* of the inner shroud **122** is not impingement cooled. Therefore, the cooling capacity of the inner shroud **122** can be suppressed to be lower than the cooling capacity of the outer shroud **121**. In this manner, the excessive cooling of the inner shroud **122** can be suppressed, the cooling of the turbine stator vane **100** can be optimized, and the turbine efficiency can be improved.

The condition (B) will be described.

For example, as shown in FIG. **2**, the above condition (B) is a condition applied in a case where the impingement plate **130** is disposed in the internal space **117**, the inner passage **171** is formed in the turbine stator vane **100**, and the inner surface **122***b* of the inner shroud **122** is impingement coolable.

As in the condition (B), in a case where the inner shroud **122** has one inner passage **171** and the cross-sectional area Si of the outlet end **171***o* of the one inner passage **171** is smaller than the total ΣSo of the cross-sectional areas So of the outlet ends **172***o* of the at least one outer passage **172**, the flow rate of the cooling medium in the inner passage **171** can be suppressed to be smaller than the total flow rate of the cooling medium in all of the outer passages **172**. Accordingly, the cooling capacity of the inner shroud can be suppressed to be lower than the cooling capacity of the outer shroud, and thus excessive cooling of the inner shroud can be suppressed and the cooling of the turbine stator vane can be optimized. Therefore, the turbine efficiency can be improved.

The condition (C) will be described.

For example, as shown in FIG. **2**, the above condition (C) is a condition applied in a case where the impingement plate **130** is disposed in the internal space **117**, the inner passage **171** is formed in the turbine stator vane **100**, and the inner surface **122***b* of the inner shroud **122** is impingement coolable.

FIG. **5** is a view of the end surface **125***c* of the outer shroud **121** on the trailing edge side in the turbine stator vane **100** shown in FIGS. **2** to **4** as seen from the downstream side.

FIG. **6** is a view of the end surface **122***c* of the inner shroud **122** on the trailing edge side in the turbine stator vane **100** shown in FIG. **2** as seen from the downstream side.

As in the condition (C), in a case where the inner shroud **122** has two or more inner passages **171** and a value (Pi/Di) obtained by dividing the interval Pi between the inner passages **171** by the diameter Di of the inner passage **171** is larger than a value (Po/Do) obtained by dividing the interval Po between the outer passages **172** by the diameter Do of the outer passage **172**, the total amount of the flow rate of the cooling medium in all of the inner passages **171** can be suppressed to be smaller than the total amount of the flow rate of the cooling medium in all of the outer passages **172**. Accordingly, the cooling capacity of the inner shroud **122** can be suppressed to be lower than the cooling capacity of the outer shroud **121**. Therefore, excessive cooling of the inner shroud **122** can be suppressed, and the cooling of the turbine stator vane **100** can be optimized. Therefore, the turbine efficiency can be improved.

That is, in some embodiments, the turbine stator vane **100** is configured to satisfy any one of the following (A), (B), or (C) conditions, and thus the cooling capacity of the inner shroud **122** can be suppressed to be lower than the cooling capacity of the outer shroud **121**. Therefore, excessive cooling of the inner shroud **122** can be suppressed, and the cooling of the turbine stator vane **100** can be optimized. Accordingly, the turbine efficiency can be improved.

In some embodiments, a value (ΣSi/(Sig+Sit)) obtained by dividing the total value ΣSi of the cross-sectional area Si of the outlet end **171***o* of the inner passage **171** by the total value of the area Sig of the gas path surface (outer surface **122***a*) of the inner shroud **122** and the area Sit of the end surface **122***c* of the inner shroud **122** on the trailing edge side may be smaller than a value (ΣSo/(Sog+Sot)) obtained by dividing the total value ΣSo of the cross-sectional areas So of the outlet ends **172***o* of the outer passage **172** by the total value of the area Sog of the gas path surface (outer surface **121***a*) of the outer shroud **121** and the area Sot of the end surface **125***c* of the outer shroud **121** on the trailing edge side.

In this case, when the areas (Sig and Sog) of the gas path surfaces (outer surfaces **121***a* and **122***a*) and the areas (Sot and Sit) of the end surfaces **125***c* and **122***c* of the shrouds **121** and **122** on the trailing edge side are compared per unit area of the total area, the flow rate of the cooling medium flowing through the inner passage **171** is smaller than the flow rate of the cooling medium flowing through the outer passage **172**. Accordingly, the cooling capacity of the inner shroud **122** can be suppressed to be lower than the cooling capacity of the outer shroud **121**. Therefore, excessive cooling of the inner shroud **122** can be suppressed, and the cooling of the turbine stator vane **100** can be optimized.

In some embodiments, a value (ΣSi**1**/Sit) obtained by dividing the total value ΣSi**1** of the cross-sectional area Si**1** of the outlet end **1710**, which is open to the end surface **122***c* of the inner shroud **122** on the trailing edge side, of the inner passage **171** by the area Sit of the end surface **122***c* of the inner shroud **122** on the trailing edge side may be smaller than a value (ΣSol/Sot) obtained by dividing the total value ΣSol of the cross-sectional areas Sol of the outlet ends **1720**, which is open to the end surface **125***c* of the outer shroud **121** on the trailing edge side, of the outer passage **172** by the area Sot of the end surface **125***c* of the outer shroud **121** on the trailing edge side.

In this case, when the flow rate of the cooling medium flowing through the inner passage **171** is compared with the flow rate of the cooling medium flowing through the outer passage **172** per unit area of the end surfaces **125***c* and **122***c* of the shrouds **121** and **122** on the trailing edge side, the flow rate of the cooling medium flowing through the inner passage **171** is smaller than the flow rate of the cooling medium flowing through the outer passage **172**. Accordingly, the cooling capacity of the inner shroud **122** can be suppressed to be lower than the cooling capacity of the outer shroud **121**. Therefore, excessive cooling of the inner shroud **122** can be suppressed, and the cooling of the turbine stator vane **100** can be optimized.

In some embodiments, an opening density of the inner impingement hole **114***i* for the impingement cooling of the inner shroud **122** may be smaller than an opening density of the outer impingement hole **114***o* for the impingement cooling of the outer shroud **121**.

Here, the opening density of the inner impingement hole **114***i* is a value obtained by dividing the total value of the opening areas of the inner impingement holes **114***i* by the area of the impingement plate **130** disposed on the inner shroud **122**.

Similarly, the opening density of the outer impingement hole **114***o* is a value obtained by dividing the total value of the opening areas of the outer impingement holes **114***o* by the area of the impingement plate **130** disposed on the outer shroud **121**.

Accordingly, the cooling capacity of the inner shroud **122** can be suppressed to be lower than the cooling capacity of the outer shroud **121**. Therefore, excessive cooling of the inner shroud **122** can be suppressed, and the cooling of the turbine stator vane **100** can be optimized.

In addition, the expression "the opening density of the inner impingement hole **114***i* for performing impingement cooling on the inner shroud **122** is smaller than the opening density of the outer impingement hole **114***o* for performing impingement cooling on the outer shroud **121**" described above also includes a case where the opening density of the inner impingement hole **114***i* for performing impingement cooling on the inner shroud **122** is zero, that is, a case where the impingement plate **130** in which the inner impingement hole **114***i* is formed is not attached as shown in FIG. **3** or FIG. **4**.

In this case, the impingement cooling of the inner shroud **122** is not performed, so that the excessive cooling of the inner shroud **122** can be suppressed and the cooling of the turbine stator vane **100** can be optimized.

(Regarding Thermal Barrier Coating)

In the turbine stator vane **100** shown in FIGS. **2** to **4**, the thermal barrier coating **180** may be applied to the airfoil portion **110**, the gas path surface (outer surface **121***a*) of the outer shroud **121**, or the like.

The construction range of the thermal barrier coating **180** in the turbine stator vane **100** shown in FIGS. **2** to **4** is, for example, the surface **110***s* of the airfoil portion **110**, the outer surface **121***a* of the outer shroud **121**, a surface **175***s* of a fillet portion **175** connecting the surface **110***s* of the airfoil portion **110** and the outer surface **121***a* of the outer shroud **121**, and the surface **175***s* of the fillet portion **175** connecting the surface **110***s* of the airfoil portion **110** and the outer surface **122***a* of the inner shroud **122** (refer to FIG. **2**).

In a case where there is a margin in the cooling capacity of the inner shroud **122**, it is conceivable to suppress the manufacturing cost of the turbine stator vane **100** by narrowing the construction range of the thermal barrier coating **180** having a relatively high construction cost.

That is, when the cooling of the inner shroud **122** by the cooling medium is suppressed as described above, the temperature of the inner shroud **122** becomes a temperature relatively close to the temperature of the combustion gas on the hub side. In such a case, the necessity of lowering the temperature of the inner shroud **122** to a temperature lower than the temperature of the combustion gas on the hub side is reduced by performing the thermal barrier coating **180** on the outer surface **122***a* of the inner shroud **122**.

Therefore, in the turbine stator vane **100** shown in FIGS. **2** to **4**, the thermal barrier coating **180** is not applied to the outer surface **122***a* of the inner shroud **122**. The thermal barrier coating **180** may be applied to a part of the outer surface **122***a* of the inner shroud **122**, not to the entire outer surface **122***a*.

Here, the outer surface **122***a* of the inner shroud **122** does not include the surface **175***s* of the fillet portion **175** that connects the surface **110***s* of the airfoil portion **110** and the outer surface **122***a* of the inner shroud **122**.

As described above, in some embodiments, a value (Sic/Sig) obtained by dividing the construction area Sic of the thermal barrier coating **180** applied to the gas path surface (the outer surface **122***a*) of the inner shroud **122** by the area Sig of the gas path surface (the outer surface **122***a*) of the inner shroud **122** may be smaller than a value (Soc/Sog) obtained by dividing the construction area Soc of the thermal barrier coating **180** applied to the gas path surface (the outer surface **121***a*) of the outer shroud **121** by the area Sog of the gas path surface (the outer surface **121***a*) of the outer shroud **121**.

In addition, the case where the value (Sic/Sig) is smaller than the value (Soc/Sog) includes a case where the construction area Sic of the thermal barrier coating **180** applied to the gas path surface (outer surface **122***a*) of the inner shroud **122** is zero.

That is, as described above, the inner shroud **122** may not be provided with the thermal barrier coating **180** on the gas path surface (outer surface **122***a*) of the inner shroud **122**.

In this way, the manufacturing cost of the turbine stator vane **100** can be suppressed by narrowing the construction range of the thermal barrier coating on the gas path surface (outer surface **122***a*) of the inner shroud **122**.

In the turbine stator vane **100** shown in FIGS. **2** to **4**, the inner shroud **122** may have the outlet passage **118** communicating with the space **110***i* inside the airfoil portion **110** and not communicating with the first recessed portion (internal space **117**).

Accordingly, in a case where the cooling medium is supplied to the serpentine channel **115** which is the space **110***i* inside the airfoil portion **110**, the cooling medium can be discharged to the outside of the turbine stator vane **100** through the outlet passage **118**.

In addition, the space **110***i* inside the airfoil portion **110** described above may be a channel of a cooling medium formed inside the airfoil portion **110** other than the serpentine channel **115**.

In some embodiments, the turbine stator vane **100** shown in FIGS. **2** to **4** may be the above-described turbine third-stage stator vane (third-stage stator vane **33**).

Accordingly, as described above, in a case where the combustion gas temperature on the hub side of the third-stage stator vane **33** is lower than the predetermined assumption and the metal temperature of the inner shroud **122** of the third-stage stator vane **33** is also lower than the predetermined assumption, the excessive cooling of the inner shroud **122** can be suppressed, and the cooling of the turbine stator vane **100** (third-stage stator vane **33**) can be optimized.

The present disclosure is not limited to the above-described embodiments, and includes a modification of the above-described embodiments and an appropriate combination of the embodiments.

For example, in some of the above-described embodiments, the outlet end **171***o* of the inner passage **171** is formed on the end surface **122***c* of the inner shroud **122** on the trailing edge side. However, at least a part of the outlet end **171***o* may be formed on the outer surface **122***a* of the inner shroud **122**.

For example, in some of the above-described embodiments, the outlet end **172***o* of the outer passage **172** is formed on the end surface **125***c* of the outer shroud **121** on the trailing edge side. However, at least a part of the outlet end **172***o* may be formed on the outer surface **121***a* of the outer shroud **121**.

For example, in some of the above-described embodiments, the outlet end **118***o* of the outlet passage **118** is formed on the end surface **122***c* of the inner shroud **122** on the trailing edge side or on the end surface **161***c* of the downstream rib **161***b* on the trailing edge side. However, at least a part of the outlet end **118***o* may be formed on the outer surface **122***a* of the inner shroud **122**.

For example, the contents described in each embodiment are understood as follows.

(1) A turbine stator vane **100** according to at least one embodiment of the present disclosure includes an airfoil

portion **110**, an inner shroud **122** that is provided on an inner peripheral side of the airfoil portion **110**, and an outer shroud **121** that is provided on an outer peripheral side of the airfoil portion **110**. The inner shroud **122** includes a first recessed portion (internal space **117**) that is formed on a surface on a side opposite to the airfoil portion **110** by interposing a gas path surface (outer surface **122***a*) of the inner shroud **122**. The outer shroud **121** includes a second recessed portion (internal space **116**) that is formed on a surface on a side opposite to the airfoil portion **110** by interposing a gas path surface (outer surface **121***a*) of the outer shroud **121**, and at least one outer passage **172** communicating with the second recessed portion (internal space **116**) and not communicating with a space **110***i* (for example, a serpentine channel **115**) inside the airfoil portion **110**. The number of the outer passages **172** is greater than the number of inner passages **171** communicating with the first recessed portion (internal space **117**) in the inner shroud **122** and not communicating with the space **110***i* (for example, the serpentine channel **115**) inside the airfoil portion **110**.

According to the configuration of (1) above, the cooling capacity of the inner shroud **122** can be suppressed to be lower than the cooling capacity of the outer shroud **121**. Therefore, excessive cooling of the inner shroud **122** can be suppressed, and the cooling of the turbine stator vane **100** can be optimized. Accordingly, the turbine efficiency can be improved.

(2) In some embodiments, in the configuration of (1) above, any one of the following conditions (A), (B), or (C) may be satisfied.

(A) The inner shroud **122** does not have the inner passage **171**.

(B) The inner shroud **122** has one inner passage **171**, and the cross-sectional area Si of the outlet end **171***o* of the one inner passage **171** is smaller than the total ΣSo of the cross-sectional areas So of the outlet ends **172***o* of the at least one outer passage **172**.

(C) The inner shroud **122** has two or more inner passages **171**, and a value (Pi/Di) obtained by dividing an interval Pi between the inner passages **171** by the diameter Di of the inner passage **171** is larger than a value (Po/Do) obtained by dividing an interval Po between the outer passages **172** by the diameter Do of the outer passage **172**.

According to the configuration of (2) above, the cooling capacity of the inner shroud **122** can be suppressed to be lower than the cooling capacity of the outer shroud **121** in a case where any one of the conditions (A), (B), and (C) is satisfied. Therefore, excessive cooling of the inner shroud **122** can be suppressed, and the cooling of the turbine stator vane **100** can be optimized. Accordingly, the turbine efficiency can be improved.

(3) In some embodiments, in the configuration of (2) above, a value (ΣSi/(Sig+Sit) obtained by dividing a total value ΣSi of the cross-sectional areas Si of the outlet ends **171***o* of the inner passages **171** by a total value of an area Sig of the gas path surface (outer surface **122***a*) of the inner shroud **122** and an area Sit of an end surface **122***c* of the inner shroud **122** on a trailing edge side may be smaller than a value (ΣSo/(Sog+Sot) obtained by dividing a total value ΣSo of the cross-sectional areas So of the outlet ends **172***o* of the outer passages **172** by a total value of an area Sog of the gas path surface (outer surface **121***a*) of the outer shroud **121** and an area Sot of an end surface **125***c* of the outer shroud **121** on a trailing edge side.

According to the configuration of (3) above, when the areas (Sog and Sig) of the gas path surfaces (outer surfaces **121***a* and **122***a*) and the areas (Sot and Sit) of the end surfaces **125***c* and **122***c* of the shrouds **121** and **122** on the trailing edge side are compared per unit area of the total area, the flow rate of the cooling medium flowing through the inner passage **171** is smaller than the flow rate of the cooling medium flowing through the outer passage **172**. Accordingly, the cooling capacity of the inner shroud **122** can be suppressed to be lower than the cooling capacity of the outer shroud **121**. Therefore, excessive cooling of the inner shroud **122** can be suppressed, and the cooling of the turbine stator vane **100** can be optimized.

(4) In some embodiments, in the configuration of (2) or (3) above, a value (ΣSi**1**/Sit) obtained by dividing a total value Si**1** of cross-sectional areas Si**1** of the outlet ends **171***o* of the inner passage **171**, which are open to an end surface **122***c* of the inner shroud **122** on a trailing edge side, of the inner passage **171** by an area Sit of the end surface **122***c* of the inner shroud **122** on the trailing edge side may be smaller than a value (ΣSol/Sot) obtained by dividing a total value ΣSol of cross-sectional areas Sol of the outlet ends **1720** of the outer passage **172**, which are open to an end surface **125***c* of the outer shroud **121** on a trailing edge side, of the outer passage **172** by an area Sot of the end surface **125***c* of the outer shroud **121** on the trailing edge side.

According to the configuration of (4) above, when the flow rate of the cooling medium flowing through the inner passage **171** is compared with the flow rate of the cooling medium flowing through the outer passage **172** per unit area of the end surfaces **125***c* and **122***c* of the shrouds **121** and **122** on the trailing edge side, the flow rate of the cooling medium flowing through the inner passage **171** is smaller than the flow rate of the cooling medium flowing through the outer passage **172**. Accordingly, the cooling capacity of the inner shroud **122** can be suppressed to be lower than the cooling capacity of the outer shroud **121**. Therefore, excessive cooling of the inner shroud **122** can be suppressed, and the cooling of the turbine stator vane **100** can be optimized.

(5) In some embodiments, in any one of the configurations of (1) to (4) above, an opening density of an inner impingement hole **114***i* for impingement cooling the inner shroud **122** may be smaller than an opening density of an outer impingement hole **114***o* for impingement cooling the outer shroud **121**.

According to the configuration of (5) above, the cooling capacity of the inner shroud **122** can be suppressed to be lower than the cooling capacity of the outer shroud **121**. Therefore, excessive cooling of the inner shroud **122** can be suppressed, and the cooling of the turbine stator vane **100** can be optimized.

(6) In some embodiments, in the configuration of (5) above, an impingement plate **130** in which the inner impingement hole **114***i* is formed is not attached to the inner shroud.

According to the configuration of (6) above, the impingement cooling of the inner shroud **122** is not performed, so that the excessive cooling of the inner shroud **122** can be suppressed and the cooling of the turbine stator vane **100** can be optimized.

(7) In some embodiments, in any one of the configurations of (1) to (6) above, a value (Sic/Sig) obtained by dividing a construction area Sic of a thermal barrier coating **180** that is applied to the gas path surface (outer surface **122***a*) of the inner shroud **122** by an area Sig of the gas path surface (outer surface **122***a*) of the inner shroud **122** may be smaller than a value (Soc/Sog) obtained by dividing a construction area Soc of a thermal barrier coating **180** that is applied to the gas

path surface (outer surface **121***a*) of the outer shroud **121** by an area Sog of the gas path surface (outer surface **121***a*) of the outer shroud **121**.

In a case where there is a margin in the cooling capacity of the inner shroud **122**, it is conceivable to suppress the manufacturing cost of the turbine stator vane **100** by narrowing the construction range of the thermal barrier coating with a relatively high construction cost.

According to the configuration of (7) above, the manufacturing cost of the turbine stator vane **100** can be suppressed by narrowing the construction range of the thermal barrier coating **180** on the gas path surface (outer surface **122***a*) of the inner shroud **122**.

(8) In some embodiments, in the configuration of (7) above, the thermal barrier coating **180** may not be applied on the gas path surface (outer surface **122***a*) of the inner shroud **122**.

In a case where there is a margin in the cooling capacity of the inner shroud, it is conceivable to suppress the manufacturing cost of the turbine stator vane **100** by narrowing the construction range of the thermal barrier coating **180** having a relatively high construction cost.

According to the configuration of (8) above, the manufacturing cost of the turbine stator vane **100** can be suppressed by not applying the thermal barrier coating **180** to the gas path surface (outer surface **122***a*) of the inner shroud **122**.

(9) In some embodiments, in any one of the configurations of (1) to (8) above, the inner shroud **122** may include an outlet passage **118** communicating with the space **110***i* (for example, a serpentine channel **115**) inside the airfoil portion **110** and not communicating with the first recessed portion (internal space **117**).

According to the configuration of (9) above, in a case where the cooling medium is supplied to the space **110***i* (for example, the serpentine channel **115**) inside the airfoil portion **110**, the cooling medium can be discharged to the outside of the turbine stator vane **100** through the outlet passage **118**.

(10) In some embodiments, in the configuration of (9) above, the space **110***i* inside the airfoil portion **110** communicating with the outlet: passage **118** may be a serpentine channel **115** for cooling the airfoil portion **110**.

According to the configuration of (10) above, the cooling medium supplied to the serpentine channel **115** for cooling the airfoil portion **110** can be discharged to the outside of the turbine stator vane **100** through the outlet passage **118**.

(11) A turbine stator vane **100** according to at least one embodiment of the present disclosure includes an airfoil portion **110**, an inner shroud **122** that is provided on an inner peripheral side of the airfoil portion **110**, and an outer shroud **121** that is provided on an outer peripheral side of the airfoil portion **110**. An opening density of an inner impingement hole **114***i* for impingement cooling the inner shroud **122** is smaller than an opening density of an outer impingement hole **114***o* for impingement cooling the outer shroud **121**.

According to the configuration of (11) above, the cooling capacity of the inner shroud **122** can be suppressed to be lower than the cooling capacity of the outer shroud **121**. Therefore, excessive cooling of the inner shroud **122** can be suppressed, and the cooling of the turbine stator vane **100** can be optimized. Accordingly, the turbine efficiency can be improved.

(12) In some embodiments, in any one of the configurations of (1) to (11) above, the turbine stator vane **100** may be a turbine third-stage stator vane (a third-stage stator vane **33**).

According to the configuration of (12) above, in a case where the combustion gas temperature on the hub side of the turbine third-stage stator vane (third-stage stator vane **33**) is lower than a predetermined assumption and the metal temperature of the inner shroud **122** of the turbine third-stage stator vane (third-stage stator vane **33**) is also lower than the predetermined assumption, the excessive cooling of the inner shroud **122** can be suppressed, and the cooling of the turbine stator vane **100** (third-stage stator vane **33**) can be optimized.

REFERENCE SIGNS LIST

1: gas turbine
33: third-stage stator vane (turbine third-stage stator vane)
100: turbine stator vane
110: airfoil portion
110*i*: space
114: through-hole
114*i*: inner impingement hole
114*o*: outer impingement hole
116: internal space (second recessed portion)
117: internal space (first recessed portion)
118: outlet passage
121: outer shroud
121*a*: outer surface
122: inner shroud
122*a*: outer surface
122*c*: end surface
125*c*: end surface
130: impingement plate
171: inner passage
171*o*: outlet end
172: outer passage
172*o*: outlet end
180: thermal barrier coating

The invention claimed is:

1. A turbine stator vane comprising:
an airfoil portion;
an inner shroud that is provided on an inner peripheral side of the airfoil portion; and
an outer shroud that is provided on an outer peripheral side of the airfoil portion,
wherein the inner shroud includes a first recessed portion that is formed on a surface on a side opposite to the airfoil portion by interposing a gas path surface of the inner shroud,
the outer shroud includes a second recessed portion that is formed on a surface on a side opposite to the airfoil portion by interposing a gas path surface of the outer shroud, and at least one outer passage communicating with the second recessed portion and not communicating with a space inside the airfoil portion,
the number of the outer passages is greater than the number of inner passages communicating with the first recessed portion in the inner shroud and not communicating with the space inside the airfoil portion,
a value (Sic/Sig) obtained by dividing a construction area Sic of a thermal barrier coating that is applied to the gas path surface of the inner shroud by an area Sig of the gas path surface of the inner shroud is smaller than a value (Soc/Sog) obtained by dividing a construction area Soc of a thermal barrier coating that is applied to the gas path surface of the outer shroud by an area Sog of the gas path surface of the outer shroud, and
the inner shroud includes an outlet passage communicating with a downstream end of a cooling flow passage

located at a most downstream side of the space inside the airfoil portion, wherein the outlet passage does not communicate with the first recessed portion.

2. The turbine stator vane according to claim **1**,

wherein the turbine stator vane satisfies either of the following conditions (A) or (B),

(A) the inner shroud includes one inner passage, and

a cross-sectional area Si of an outlet end of the one inner passage is smaller than a total ΣSo of cross-sectional areas So of outlet ends of the at least one outer passage,

(B) the inner shroud includes two or more inner passages, and

a value (Pi/Di) obtained by dividing an interval Pi between the inner passages by a diameter Di of the inner passage is larger than a value (Po/Do) obtained by dividing an interval Po between the outer passages by a diameter Do of the outer passage.

3. The turbine stator vane according to claim **2**,

wherein a value (ΣSi/(Sig+Sit)) obtained by dividing a total value ΣSi of the cross-sectional areas Si of the outlet ends of the inner passages by a total value of an area Sig of the gas path surface of the inner shroud and an area Sit of an end surface of the inner shroud on a trailing edge side is smaller than a value (ΣSo/(Sog+Sot)) obtained by dividing a total value ΣSo of the cross-sectional areas So of the outlet ends of the outer passages by a total value of an area Sog of the gas path surface of the outer shroud and an area Sot of an end surface of the outer shroud on a trailing edge side.

4. The turbine stator vane according to claim **2**,

wherein a value (ΣSi**1**/Sit) obtained by dividing a total value ΣSi**1** of cross-sectional areas Si**1** of the outlet ends of the inner passage, which are open to an end surface of the inner shroud on a trailing edge side, of the outlet ends of the inner passage by an area Sit of the end surface of the inner shroud on the trailing edge side is smaller than a value (ΣSol/Sot) obtained by dividing a total value ΣSol of cross-sectional areas Sol of the outlet ends of the outer passage, which are open to an end surface of the outer shroud on a trailing edge side, of the outlet ends of the outer passage by an area Sot of the end surface of the outer shroud on the trailing edge side.

5. The turbine stator vane according to claim **1**,

wherein an opening density of an inner impingement hole for impingement cooling the inner shroud is smaller than an opening density of an outer impingement hole for impingement cooling the outer shroud.

6. The turbine stator vane according to claim **5**,

wherein an impingement plate in which the inner impingement hole is formed is not attached to the inner shroud.

7. The turbine stator vane according to claim **1**,

wherein the thermal barrier coating is not applied to the gas path surface of the inner shroud.

8. The turbine stator vane according to claim **1**,

wherein the space inside the airfoil portion communicating with the outlet passage is a serpentine channel for cooling the airfoil portion, and

the outlet passage includes an inlet opening formed in a wall surface located at a most downstream end, with respect to a flow direction of a cooling medium flowing through the serpentine channel, of wall surfaces defining the serpentine channel.

9. The turbine stator vane according to claim **1**,

wherein the turbine stator vane is a turbine third-stage stator vane.

10. The turbine stator vane according to claim **1**, wherein the inner shroud does not include the inner passage.

11. A turbine stator vane comprising:

an airfoil portion;

an inner shroud that is provided on an inner peripheral side of the airfoil portion; and

an outer shroud that is provided on an outer peripheral side of the airfoil portion,

wherein an opening density of an inner impingement hole for impingement cooling the inner shroud is smaller than an opening density of an outer impingement hole for impingement cooling the outer shroud,

a value (Sic/Sig) obtained by dividing a construction area Sic of a thermal barrier coating that is applied to the gas path surface of the inner shroud by an area Sig of the gas path surface of the inner shroud is smaller than a value (Soc/Sog) obtained by dividing a construction area Soc of a thermal barrier coating that is applied to the gas path surface of the outer shroud by an area Sog of the gas path surface of the outer shroud, and

the inner shroud includes an outlet passage communicating with a downstream end of a cooling flow passage located at a most downstream side of the space inside the airfoil portion, wherein the outlet passage does not communicate with the first recessed portion.

12. The turbine stator vane according to claim **11**,

wherein the thermal barrier coating is not applied to the gas path surface of the inner shroud.

13. The turbine stator vane according to claim **11**,

wherein the space inside the airfoil portion communicating with the outlet passage is a serpentine channel for cooling the airfoil portion, and

the outlet passage includes an inlet opening formed in a wall surface located at a most downstream end, with respect to a flow direction of a cooling medium flowing through the serpentine channel, of wall surfaces defining the serpentine channel.

* * * * *